

Name: Trina Murrell

60

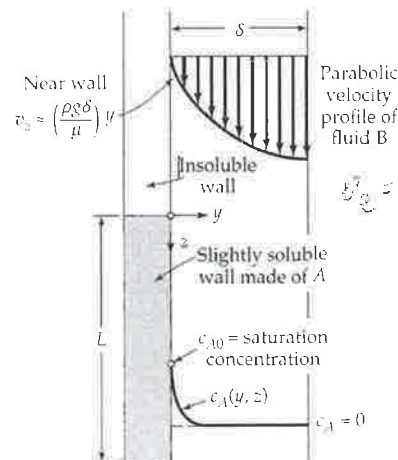
TRANSPORT PHENOMENA II, CHE 312

QUIZ 4

CLOSED BOOK, TIME: 20 MINUTES

Liquid B flows in laminar motion down a vertical wall, with the velocity profile

$$v_z(y) = \frac{\rho g \delta^2}{2\mu} \left[ 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \right]$$



For  $0 < z < L$  the wall is coated with a species A that is slightly soluble in B ( $x_A \ll 1$ ). For short distances downstream, species A will not diffuse very far into the falling film. That is, A will be present only in a very thin boundary layer near the solid surface.

Write a shell mole balance for species A within the liquid film and derive a partial differential equation for the concentration of A as a function of y and z.

Hint: combined mole flux is  $\langle N_A \rangle = c_A \langle v^* \rangle + \langle J_A^* \rangle$ , where the diffusive flux according to Fick's law is  $\langle J_A^* \rangle = -c D_{AB} \langle \nabla x_A \rangle$ .

$$0 = [A N_{Az}]_z - [A N_{Az}]_{z+\Delta z} - [A N_{Ay}]_y - [A N_{Ay}]_{y+\Delta y}$$

$$0 = \frac{\partial N_z}{\partial y \Delta z} \approx \frac{\partial N_z}{\partial z} - \frac{\partial N_y}{\partial y}$$

$$0 = c_A \left( \frac{\rho g \delta^2}{2\mu} \right) \left[ 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \right] - c D_{AB}$$

$$\frac{c D_{AB}}{\left( \frac{\rho g \delta^2}{2\mu} \right) \left[ 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \right]} = c_A$$

\* phys. int. ok  
\* incomplete

$$\langle N_A \rangle = c_A \langle v^* \rangle + \langle J_A^* \rangle$$

$$\langle J_A^* \rangle = -c D_{AB} \langle \nabla x_A \rangle$$

$$v_z(y) = \frac{\rho g \delta^2}{2\mu} \left[ 2\left(\frac{y}{\delta}\right) - \left(\frac{y}{\delta}\right)^2 \right]$$

$$\checkmark N_{Az} = c_A \langle v_z \rangle + \langle J_{Az}^* \rangle$$

$$\checkmark N_{Ay} = c_A \langle v_y \rangle + \langle J_{Ay}^* \rangle$$

$$N_{Az} = c_A \langle v_z(y) \rangle - c D_{AB} \langle \nabla x_A \rangle$$